Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*	(i)		moles CaO = $\frac{1.90}{56}$ = 0.0339 (1) moles HCl = $\frac{50 \times 1.40}{1000}$ = 0.070 (1) 0.07 mol HCl would neutralise 0.035 mol CaO so acid in excess (1)		3		3	1 1	3
		(ii)		4284.5		1		1		
		(iii)		$\frac{mc\Delta T}{n}$ / $\frac{4284.5}{0.0339}$ (1) -126.4 kJ mol^{-1} **both** sign and value needed (1)		2		2	1	
		(iv)		Hess diagram shown with arrows in correct direction (1) ignore products of reactions $\Delta_r H$ = 126.4 – 196 = –69.6 kJ mol^{-1} (1) *(using value given in question 110 – 196 = –86 kJ mol^{-1})*		2		2	1	
		(v)		award (1) each for any two of following • suitable apparatus to minimise **heat losses** e.g. lid/ polystyrene container • thermometer reading to 0.1°C / graduations to allow reading to less than 0.5°C • use a burette since it can be read to 0.05 cm^3			2	2		2